



**WEAPONS
ROBOTICS AND
CONTROL
ENGINEERING**
UNITED STATES NAVAL ACADEMY

HOPPER INFORMATION SERIES

05 / BINARY COMPUTER VISION

Thresholding with Hopper

Turn camera brightness into a reliable two-color signal for a white-paper mission.

WHITE + BLACK

0-100% THRESHOLD

DOWNWARD CAMERA





One camera frame becomes two colors



CAMERA -> LUMINANCE -> THRESHOLD RULE -> WHITE / BLACK

One threshold turns a grayscale frame into a mask

1 · CONVERT COLOR TO LUMINANCE

$$Y(x, y) = 0.2126R + 0.7152G + 0.0722B$$

2 · APPLY THE SLIDER THRESHOLD

$$B(x, y) = \begin{cases} 1, & Y(x, y) \geq 2.55T \\ 0, & Y(x, y) < 2.55T \end{cases}$$

3 · MEASURE WHITE COVERAGE

$$C_{\text{white}} = \frac{100}{WH} \sum_{x=1}^W \sum_{y=1}^H B(x, y)$$

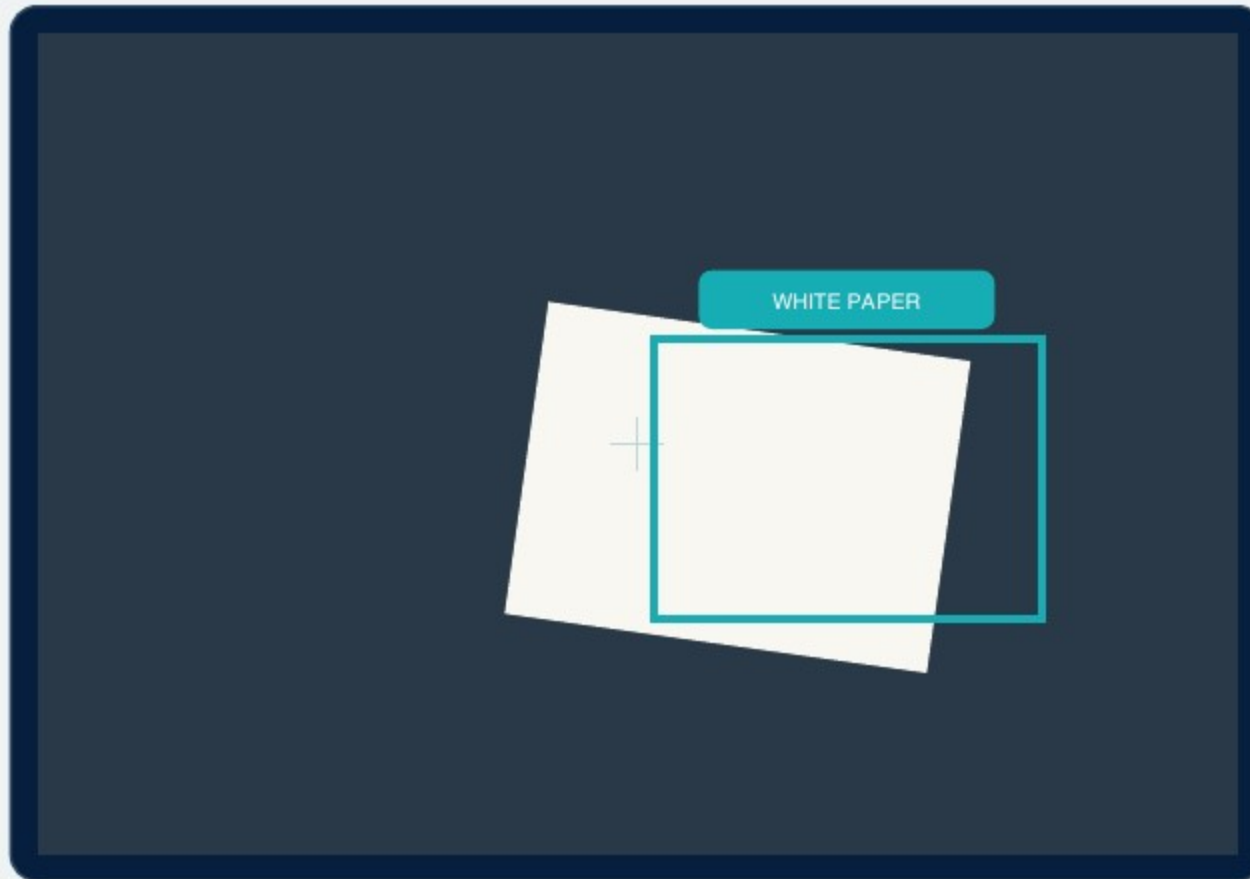
How to read it

- T is the 0–100 slider value; 2.55T maps it to an 8-bit intensity.
- B = 1 means white in the binary mask; B = 0 means black.
- Invert swaps the classes: $B_{\text{inv}} = 1 - B$.

Thresholding uses a rule you can inspect—not a learned neural network.



A white sheet can become a visual landing cue



CALIBRATE BEFORE FLIGHT

- Place the real paper on the real floor.
- Use the exact classroom lighting.
- Adjust T until paper stays white and floor stays black.
- Record the chosen threshold.

CHOOSE A DECISION

- Coverage: enough of the frame is paper.
- Center pixel: reticle has crossed onto paper.
- Combine repeated checks for a safer decision.
- Then explicitly command land.

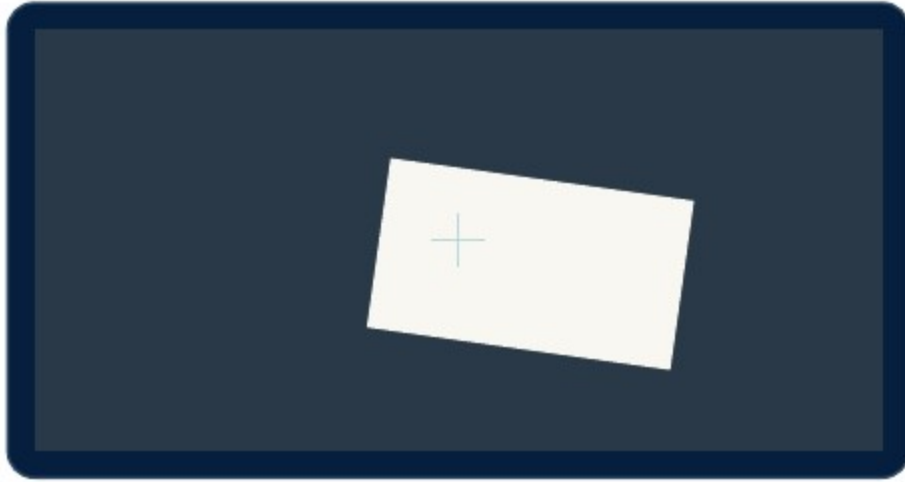
Thresholding detects brightness, not the semantic concept 'paper'.



Two useful sensing strategies

FRAME COVERAGE

Ask: what percentage of all pixels are white?

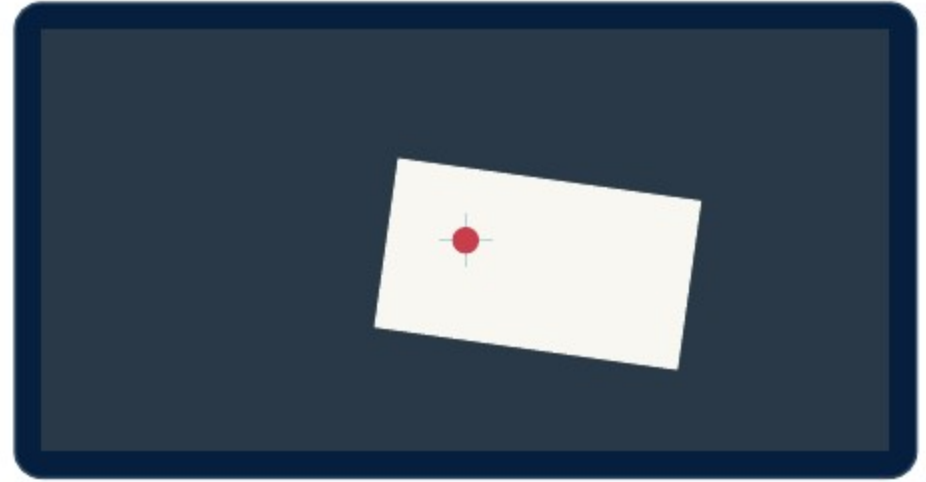


```
whiteCoverage >= 18%
```

Good for: area entered, target fills view

CENTER PIXEL

Ask: is the single reticle pixel white?

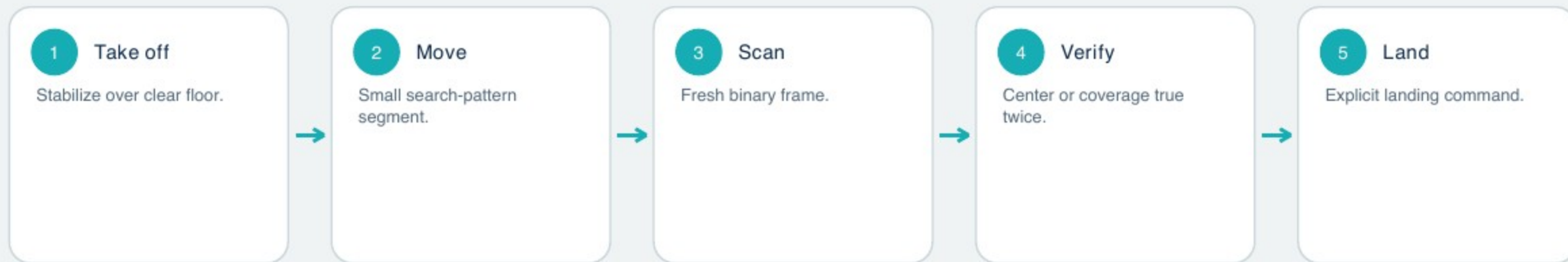


```
centerWhite === true
```

Good for: center over target, line crossing



Search, verify, then land



SEARCH CHECK

```
1  const hit = await
2    vision.seesBinary("white", 60, false, 18);
3  if (hit) {
4    await drone.hover();
5    await drone.land();
6  }
```

SAFER THAN ONE CHECK

- Move in small increments.
- Pause for exposure and motion to settle.
- Require repeated consistent observations.
- Keep a timeout and fallback land.



Lighting is part of the experiment

SHADOW

A shadow can turn white paper gray or black.

REFLECTION

Glare can turn dark floor pixels bright.

AUTO-EXPOSURE

The same paper may shift brightness as the view changes.

MOTION BLUR

Fast movement mixes paper and floor at the edge.

FLOOR TEXTURE

Light markings can resemble the target.

TARGET SIZE

A small target may never reach the coverage threshold.

SOURCES

Hopper Studio lib/vision.ts · FTW Robotics indoor-flight guidance